

MUHAMMAD USMAN OMER

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EDUCATION

University of British Columbia

University of British Columbia Vancouver, BC

Vancouver, BC

Vancouver, BC

Bachelor of Science in Computer Science

Sep. 2023

- Dean's List (2024)

EXPERIENCE

Machine Learning Intern

Aug 2021

Punjab Information Technology Board

Lahore, Pakistan

- Developed a machine learning model to track illegal license plates via CCTV using R, improving detection accuracy by 25%.
- Collaborated in a team to implement and test the model in a real-world setting, ensuring scalability and efficient performance.

Machine Learning Intern

Jun 2022

Dubai, UAE

Himmah Capital Limited

Himmah Capital Limited Dubai, UA

- Applied machine learning techniques to analyze trends in the Saudi Stock Exchange (TASI), contributing to a 127% enhancement in prediction accuracy through KNN and big data analytics.
- Developed and maintained scalable data pipelines for efficient data processing and analysis.

PROJECTS

Ticket Booking Application | Java, Spring Boot, Spring Security, React, Vite

- Designed and implemented a robust, scalable ticket booking platform with secure login and registration functionality, featuring a modern, responsive UI built with React and Vite.
- Utilized Spring Data JPA to streamline CRUD operations for managing user bookings, cancellations, and updates, ensuring seamless interaction with the backend database and improved performance by 30%.
- Implemented Spring Security and JWT-based authentication to safeguard user data and restrict access to authorized endpoints, enhancing overall application security and reducing vulnerabilities by 15%.
- Developed RESTful API endpoints using Spring Boot REST controllers, enabling efficient communication between the frontend and backend.
- Introduced a well-structured service layer to handle business logic, ensuring scalability, maintainability, and separation of concerns within the application.
- Backend Repository | Frontend Repository | Live Application

FUT Team Builder Application | Java, Swing, JUnit, Mockito

Swing, JU

- Developed a comprehensive application for FIFA Ultimate Team (FUT) enthusiasts to create, manage, and optimize custom teams with an interactive gaming-style interface using Java Swing.
- Implemented secure login and registration functionality using Java's UserManager, ensuring role-based access control for users and admins.

- **DESIGNED A COMPLEX SERVICE LAYER WITH ALGORITHMS TO CALCULATE KEY TEAM METRICS SUCH AS CHEMISTRY, AVERAGE RATING, AND TOTAL PRICE, IMPROVING THE USER EXPERIENCE AND REDUCING CALCULATION TIME BY 20%.**

SFTWR Labs Web Application | Java, Spring Boot, Next.js

- Integrated a community feature that allows users to save and share their teams in a public repository, enabling others to browse, filter, and like popular team builds.
- Achieved 90% code coverage through comprehensive unit testing using JUnit and Mockito, implementing test cases for all service layer components and user interactions.
- Source Code
- Designed and developed a full-stack business application for SFTWR Labs, featuring a secure backend built with Spring Boot and a dynamic frontend using Next.js.
- Implemented a RESTful API architecture and utilized Spring Data JPA for CRUD operations, managing data through Spring REST endpoints, resulting in a 25% increase in data processing efficiency.
- Developed an Admin Panel with Spring Security and integrated Slack API for automated notifications and Spring Mail for email communications, improving team communication and reducing response time by 10%.
- Visit SFTWR Labs | Source Code (Backend) | Source Code (Frontend)

TECHNICAL SKILLS

Languages : Java, C, C++, Python, SQL, JavaScript, HTML/CSS

Python, SQ

Frameworks : Spring Boot, Spring Security, Spring Data JPA, React, Hibernate, Next.js

Tools : Linux, Git, Redis Cache, Amazon Bedrock, Open AI API, Apache Kafka

DATABASES: POSTGRESQL (EXPERIENCE FROM PROJECTS)
